

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Backend Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built an AI-assisted workflow for structured company-registration drafts, combining prompt outputs, operator validation, public-site API lookups, retries, and automation handoff.
- Implemented async backend execution with browser automation, session recovery, state validation, and failure-aware retries, reducing manual filing time by **85%**.
- Containerized services with Docker, deployed to Google Cloud Run, and automated releases through GitHub Actions, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python medical-image segmentation tooling across **95 brain subparts**, covering preprocessing, orientation correction, model inference, generated masks, and large MRI data paths.
- Reduced segmentation runtime by **87%** using asynchronous and parallel execution while preserving inspection workflows for generated masks and 3D surfaces.
- Debugged model-output pipeline issues across conformation, label mapping, PyTorch/ONNX-style inference, and binary mask generation inside a mature open-source codebase.

Open Source

[Contributions](#) 

- Contributed to **Kuadrant MCP Gateway** reliability behavior, **Kyverno** policy/infrastructure paths, and IOOS/Tiled data or visualization tooling under maintainer review.

Projects

Hyoka | *Python, FastAPI, SQLAlchemy, Alembic, PostgreSQL, OpenTelemetry, Next.js, Docker*

- Built an **AI-agent reliability control plane** with SDK/proxy/OTLP trace ingestion, evaluator registries, replay modes, failure mining, validation runs, gate decisions, promotions, and audit logs.
- Implemented worker-owned execution leases, content-addressed artifacts, signed manifests, project-scoped API keys, SQLAlchemy/Alembic models, and a Next.js dashboard for operating agent behavior safely.

Swish Support System | *JavaScript, React, Python, PostgreSQL/pgvector, Redis, Langfuse*

[GitHub](#) 

- Built a hybrid AI + rules support system where LLM assessments classify messy complaint semantics while deterministic policy code controls refunds, replacements, coupons, and escalations.
- Grounded responses in order, item, evidence, customer, and policy context with PostgreSQL/pgvector, Redis state, Langfuse traces, confidence thresholds, and regression checks.

Penny Lane Capital | *Python, Agno, Pydantic, SQLite, Vector Memory, yfinance, JSONL*

[GitHub](#) 

- Built a multi-agent market-research workflow with analyst roles, bull/bear debate, trader, risk review, portfolio approval, typed Pydantic state, SQLite checkpoints, and replayable JSONL traces.
- Evaluated decisions with historical-market backtests, reward-loop style scorecards, validation harnesses, and baseline comparisons across return, drawdown, win-rate, and risk signals.

Technical Skills

Languages: Python, TypeScript, Go, SQL, C++

Frameworks: FastAPI, PyTorch, React, Next.js, Node.js, Agno, VTK, wxPython

Tools: PostgreSQL/pgvector, Redis, SQLite, Vector Memory, OpenTelemetry, Docker, GCP, GitHub Actions, Langfuse, ONNX